

Chrystal Chern

Dynamics Lab / STAIRLab / Berkeley Discovery

University of California, Berkeley

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Dynamics researcher specializing in IMU-based motion sensing, system identification, and digital twins of physical systems, with experience spanning sensor instrumentation, standards-based experimental testing, and signal-level characterization through production-quality Python analysis tools.

Education

- 2024 **PhD | Structural Engineering, Mechanics, and Materials, University of California, Berkeley,**
GPA: 3.89/4.00.
Thesis: *Digital Twin Framework for Vibration-Based Structural Health Monitoring*
Advisor: Khalid M. Mosalam
Major Focus: Structural mechanics and dynamics, linear and nonlinear structural analysis, finite element methods, performance-based engineering and design.
Minor Focus: Artificial intelligence, machine learning, control systems, and scientific computing.
- 2019 **MS | Structural Engineering, Mechanics, and Materials, University of California, Berkeley.**
Thesis: *Deep Learning for Transmission Tower Structural Health Monitoring in Small Datasets*
- 2016 **BS | Mechanical Engineering, Massachusetts Institute of Technology, Cambridge.**
Mechanical systems, thermodynamics, and design.

Research Experience

University of California, Berkeley

- 2024–Present **Postdoctoral Researcher, Dynamics Lab & STAIRLab, Berkeley, CA.**
Dynamics Lab:
- Lead instrumentation, signal processing, and inverse dynamic analysis of coupled human-object biomechanical systems using multi-node IMU sensor networks with strap-down integration and magnetic field mapping; led experiment design, sensor placement, and data collection.
 - Developed network analysis methodology and Python analysis code to identify the causal body-segment motions driving induced object motion (*Nonlinear Dynamics*, submitted 2026).
- STAIRLab (STructural Artificial Intelligence Research Lab):**
- Build finite element models to generate analytical benchmarks, validating system-identification methods across experimental, field, and FEA-simulated structural response data.
 - Evaluate system identification prediction residuals and modal clustering for anomaly detection and early-warning structural health monitoring under environmental and seismic loads.
 - Supervise graduate and undergraduate researchers in data-driven system dynamics.
- 2019–2024 **Graduate Student Researcher, STAIRLab & Pacific Earthquake Engineering Research Center (PEER), Berkeley, CA.**
Bridge Rapid Assessment Center for Extreme Events (BRACE2):
- Developed California's production web platform for automated bridge health monitoring (Caltrans).
 - Performed system identification and finite element modeling of bridges and buildings.
- Structural ImageNet & Feature Engineering for Structural Health Monitoring:**
- Developed machine learning anomaly detection for bridge damage diagnosis within a human-machine collaboration framework (*Journal of Bridge Engineering*, 2024).
 - Developed deep learning image classification models for structural health monitoring.
 - Built feature engineering pipelines for damage classification.

Massachusetts Institute of Technology

2015–2016 **Undergraduate Researcher**, *MIT Digital Structures Lab*, Cambridge, MA.
Experimental characterization of fine-resolution anisotropic material properties of 3D printed materials via ASTM tensile testing.

Professional Engineering Experience

2016–2018 **Staff I – Building Technology**, *Simpson Gumpertz & Heger Inc.*, Washington, DC.

- Performed parametric energy, daylighting, and thermal finite element analysis (2D/3D) of curtain wall, cladding, and custom skylight assemblies.
- Conducted infrared thermography surveys and analysis for non-destructive diagnostics.
- Performed on-site water penetration and air infiltration testing of installed cladding assemblies per ASTM, AAMA, and ASHRAE standards, including instrumented pressure-chamber test setup, acceptance-criteria evaluation, and leak-path diagnosis.
- Analyzed wind loads on components and cladding per structural code.

2014 **Engineering Intern**, *Dassault Systemes SOLIDWORKS Corp.*, Woodland Hills, CA.

- Investigated user-reported software issues across Computer Aided Design (CAD) and Finite Element Analysis (FEA) products for the SOLIDWORKS 2015 Beta program.
- Contributed to the redesign of the engineering technical support knowledge base.

Technical & Computer Skills

Programming (production)	Python (package/module development), Bash, Tcl, HTML/CSS, JavaScript
Additional Languages & Tools	MATLAB, R, C, Rust
Modeling & Analysis	Vibrations and structural dynamics, differential equation modeling, linear algebra, numerical methods, finite element analysis of material and structural behavior (mechanical and thermal), machine learning anomaly detection
Test & Measurement	Data acquisition (DAQ), sensor instrumentation, signal processing (FFT, Kalman filtering, strap-down integration, system identification), magnetic field mapping, ASTM standards-based testing (tensile, water penetration, air infiltration), infrared thermography, field acceptance testing
Languages	English (Native), Mandarin Chinese, Spanish

Fellowships & Awards

2024–Present **UC Berkeley Future of Higher Education Postdoctoral Fellowship** (\$140k)

2020–2024 **NSF Graduate Research Fellowship (GRFP)** (\$160k)

2015 **MIT Priscilla King Gray Public Service Center Expedition Grant** (\$10k)

2012 **Intel Scholarship** (\$4k)

Publications

Journal Articles

2026 **Chrystal Chern** and Oliver M. O'Reilly. "On Complex Modal Inverse Analysis: The Cautionary Tale of the Single Surviving Mode." *ASME Journal of Applied Mechanics*, 2026. <https://doi.org/10.1115/1.4072378>

- Submitted **Chrystal Chern**, Theresa E. Honein, and Oliver M. O'Reilly. "The Symphony of Gyration Producing Steady Motions of a Hula Hoop: Insights from a Network Analysis." *Nonlinear Dynamics*. Submitted May 2026.
- In Prep. **Chrystal Chern** and Khalid Mosalam. "Modal Clustering for Digital Twinning of Civil Infrastructure." *Journal of Engineering Mechanics*.
- In Prep. Naiqi Guo, **Chrystal Chern**, and Khalid Mosalam. "Detecting Inelasticity in Seismic Data with Time Response System Identification." *Mechanical Systems and Signal Processing*.
- 2024 Sifat Muin, **Chrystal Chern**, and Khalid Mosalam. "Human-Machine Collaboration Framework for Bridge Health Monitoring." *Journal of Bridge Engineering*, 29(7), 04024041. <https://doi.org/10.1061/JBENF2.BEENG-6587>

Conference Proceedings

- 2025 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. "Structural Response Prediction from Learned System Realization Matrices." *11th International Conference on Experimental Vibration Analysis of Civil Engineering Structures (EVACES)*. https://doi.org/10.1007/978-3-031-96110-6_72
- 2024 **Chrystal Chern** and Khalid Mosalam. "Cross-Sectional Study of Physics-Informed Bridge Health Identification." *International Association of Bridge Earthquake Engineering (IABEE) Fourth International Bridge Seismic Workshop (4IBSW)*.
- 2020 Sifat Muin, **Chrystal Chern**, and Khalid Mosalam. "Human-Machine Collaboration Framework for Bridge Health Monitoring." *SMIP20 Seminar on Utilization of Strong-Motion Data Proceedings*, pp. 100-127.

Software

- 2024 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. mdof: 0.0.16-alpha, open-source package for system identification. Zenodo. <https://doi.org/10.5281/zenodo.11660201>

Technical Reports & Dissertations

- 2024 **Chrystal Chern**. "Digital Twin Framework for Vibration-Based Structural Health Monitoring." *PhD Dissertation*, UC Berkeley.
- 2024 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. "BRACE2: Bridge rapid assessment center for extreme events, Phase I Final Report." *Caltrans Technical Report No. CA24-3703*.
- 2019 **Chrystal Chern**. "Deep Learning for Transmission Tower Structural Health Monitoring in Small Datasets." *MS Research Report*.

Teaching & Mentorship

- Spring 2026 **Lecturer (Instructor of Record)**, L&S 110 – *Brilliance of Berkeley* (UC Berkeley). Advise students, prepare course materials, grade, facilitate discussions, and coordinate guest speakers; supervise student readers.
- 2024–2025 **Teaching Assistant**, MAS-E ENGIN 235-A – *Python for Engineers* (UC Berkeley). Authored programming laboratory assignments, developed autograder scripts, and reviewed lectures and quizzes.
- Dec 2024 **Guest Lecturer**, CE 249 – *Experimental Methods in Structural Engineering* (UC Berkeley). Delivered a lecture on system identification of vibration data using computer vision.

- 2019, **Graduate Student Instructor, E7 – Computer Programming for Scientists and Engineers** (UC Berkeley).
Delivered discussion lectures, worksheets, and laboratory assignments; led exam review sessions and graded exams.
- 2025 **Lead Organizer & Founder, SEMM Graduate Program Primer Boot Camp**, UC Berkeley.
Founded and taught a primer on core theory and field-specific conventions for incoming graduate students; organized 7 doctoral co-instructors.

Service & Leadership

- 2024–Present **Program Strategist**, Berkeley Discovery Initiative, UC Berkeley – Co-developed an interdisciplinary research mentorship program (8 research hubs, 30+ graduate fellows, 170 undergraduates); led survey and focus-group assessment and co-secured \$170k in continuing funding.
- 2022–Present **Technical Reviewer** for *Resilient Cities and Structures*, *Computers and Structures*, and *Computer-Aided Civil and Infrastructure Engineering*.
- 2024–Present **Research Mentor**, UC Berkeley SEMM & CEE Scholars Programs – Designed and supervised structural health monitoring and sensor projects for 5 undergraduate researchers.
- 2022–2024 **Founder & President**, UC Berkeley SEMM-DSR (Doctoral Students and Researchers Organization).
- 2014–2015 **Malawi Project Manager & Fundraising Chair**, MIT Engineers Without Borders.

References

Oliver M. O'Reilly

Postdoctoral Research Supervisor

Distinguished Professor of Mechanical Engineering, UC Berkeley

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Khalid M. Mosalam

PhD Thesis Advisor & Postdoctoral Supervisor

Taisei Professor of Civil Engineering & Director, PEER Center, UC Berkeley

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Leslie R. Harlson

Postdoctoral Supervisor

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